UNITED INTERNATIONAL UNIVERSITY
School of Business and Economics (SoBE)

Course Outline: BSA 4201 – Applied Decision Modelling

Program: BBA
Trimester: Fall 2026
Course Title: Applied Decision Modelling
Course Code: BSA 4201
Course Type: Business Analytics Major Course
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Office Location: *Room 219 (C),* UIU Main Campus, United City, Satarkul, Badda, Dhaka
Pre-requisites: CST 2321
Credit Hours: 3
Weekly Hours: 3
Total Marks: 100

Course Rationale

This course builds hands-on decision-modelling skills in Python (Jupyter Notebook), from data preparation and visualization to decision analysis, optimization, and predictive analytics, enabling data-driven business decisions.

Course Objectives

1. Explain the role of analytics and decision modelling in business decision-making.
2. Use Python in Jupyter Notebook, including Pandas, to clean, transform, and summarize business data.
3. Create meaningful visualizations with Matplotlib and Seaborn to explore data and communicate insights.
4. Structure and evaluate decisions under certainty, uncertainty, and risk using deterministic models, decision criteria, decision trees, and Bayesian revision.
5. Formulate and solve network optimization and linear programming models for business problems using Python.
6. Build, evaluate, and interpret predictive models (logistic regression, boosting) and basic NLP applications, and present findings through an applied analytics project.

Course Content Overview

- **Python and Jupyter Foundations:** Jupyter Notebook environment, Python essentials, and data wrangling with Pandas
- **Data Visualization:** Exploratory data analysis and visual storytelling with Matplotlib and Seaborn
- **Decision Modelling:** Deterministic models; decisions under uncertainty (Maximax, Maximin, Laplace, Hurwicz); decision trees; Bayesian revision and EVSI
- **Optimization Models:** Network optimization and linear programming formulated and solved in Python
- **Predictive Analytics:** Logistic regression and boosting models (AdaBoost, XGBoost, LightGBM, CatBoost)
- **Text Analytics:** Introduction to Natural Language Processing (NLP) for business data
- **Applied Analytics Project:** Staged chapter submissions, final report, and presentation

Course Progression Planning

Class	Topic
1	Introduction/Basic Knowledge about Jupyter Notebook
2-3	Pandas
4	Visualization with Matplotlib and Seaborn
5	Building a deterministic model
6	Decisions under Uncertainty (Maixmax, maximin, Laplace, Hurwicz)
7	Quiz 1
8	Decision Tree
9	Bayesian Revision & EVSI
10	Network Optimization
11	Quiz 2
12	Linear Programming
Mid Term Examination	
13	Logistics Regression
14	Boosting models: AdaBoost, XGBoost, LightGBM, and CatBoost
15	NLP
16	Quiz 3
17-18	Chapter 1 & 2 Submission of Project
19-20	Chapter 3 & 4 Submission
21-22	Chapter 5 & 6 Submission
23-24	Final Project Submission & Presentation

CLOs and PLO Mapping

CLO	Description	Mapped PLOs
1	Explain the role of analytics and decision modelling in business, and use Python in Jupyter Notebook (Pandas) to clean, transform, and summarize business data	PLO1, PLO4
2	Create and interpret data visualizations with Matplotlib and Seaborn to explore patterns and communicate business insights	PLO2, PLO4
3	Analyze and evaluate business decisions under certainty, uncertainty, and risk using deterministic models, decision criteria, decision trees, Bayesian revision, and EVSI	PLO1, PLO2
4	Formulate and solve network optimization and linear programming models for business problems using Python	PLO2, PLO4
5	Build, evaluate, and interpret predictive (logistic regression, boosting) and NLP models, and present findings through a team-based analytics project with responsible and ethical use of data	PLO2, PLO3, PLO4, PLO5

PLO1: Knowledge and understanding, PLO2: Critical Judgement, Problem Solving and Numeracy Skills

PLO3: Interpersonal Skills, PLO4: IT Skills, PLO5: Ethical, Social and Environmental Accountability

Teaching-Learning & Assessment Strategy Summary

Topic	Teaching Methods	Assessments
Python and Jupyter Foundations (Pandas)	Lectures, Live Coding Demonstrations in Jupyter Notebook, Guided Lab Exercises	Class Tasks, Quiz 1, Mid-Term (CLO1)
Data Visualization (Matplotlib, Seaborn)	Hands-on Lab Sessions, Real-life Datasets, Discussions	Notebook Assignments, Quiz 1 (CLO2)
Decision Modelling (Deterministic Models, Decisions under Uncertainty, Decision Trees, Bayesian Revision & EVSI)	Lectures, Worked Business Cases, Python Implementation	Quiz 1, Quiz 2, Mid-Term (CLO3)
Optimization Models (Network Optimization, Linear Programming)	Lectures, Problem-solving Workshops, Coding Labs with Python Solvers	Quiz 2, Mid-Term, Notebook Assignments (CLO4)
Predictive Analytics (Logistic Regression, Boosting Models)	Hands-on Labs, Model Building and Evaluation in Python, Case Studies	Quiz 3, Project Chapters (CLO5)
Text Analytics (NLP)	Demonstrations, Guided Labs on Business Text Data	Quiz 3, Project Chapters (CLO5)
Applied Analytics Project	Team-based Project Work, Instructor Consultations, Peer Feedback	Staged Chapter Submissions, Final Report and Presentation (CLO1–CLO5)

Assessment and Grading

Component	Marks
Attendance and Participation	**5%**
Quizzes/Class Tests (top 2 out of 3)	**15%**
Term Paper/Project	**60%**
Mid-Term	**20%**
Total	**100%**

Grading Policy

Letter	Marks	Grade Point	Letter	Marks	Grade Point
A (Plain)	90–100	4.00	C (Plain)	66–69	2.00
A− (Minus)	86–89	3.67	C− (Minus)	62–65	1.67
B+ (Plus)	82–85	3.33	D+ (Plus)	58–61	1.33
B (Plain)	78–81	3.00	D (Plain)	55–57	1.00
B− (Minus)	74–77	2.67	F	<55	0.00
C+ (Plus)	70–73	2.33			